

KAIROTHON 2K26

Civil Engineering

KH26-CE-21 – AI-Based Urban Flood Risk Prediction and Evacuation Planning System

Background

Urban flooding is increasing because of heavy rainfall, blocked drainage systems, rapid construction and poor land-use planning. Authorities often lack timely information about vulnerable locations and safe evacuation routes.

Problem Statement

Develop an intelligent platform that analyses rainfall, drainage capacity, elevation, water-level and historical flood information to identify high-risk areas. The system should display risk zones on a map and recommend safe evacuation routes and temporary shelter locations. It should update the risk level when environmental conditions change and generate alerts for affected communities. Authorities must be able to monitor blocked routes, shelter capacity and emergency-response status.

Mandatory Requirements

- Rainfall and water-level data analysis
- AI-based flood-risk classification
- Map-based risk-zone visualization
- Evacuation-route recommendation
- Shelter-capacity monitoring
- Public and administrative alerts

Expected Prototype

A map-based dashboard demonstrating flood-risk prediction and evacuation planning using simulated or publicly available data.

KH26-CE-22 – AI-Assisted Structural Health Monitoring and Maintenance Prioritization System

Background

Bridges and buildings may develop cracks, excessive vibration, corrosion or structural deformation over time. Manual inspection is expensive and may not identify early signs of deterioration.

Problem Statement

Develop an AI-supported structural-health monitoring platform that analyses images, vibration readings, strain data or inspection records. The system should identify possible defects, estimate their severity and calculate a structural-health score. It should prioritize structures requiring immediate inspection or maintenance and maintain a history of detected problems. Engineers should be able to compare current and previous inspection results through a dashboard.

Mandatory Requirements

- Image or sensor-data analysis
- Crack or defect identification
- Structural-health scoring
- Severity classification
- Maintenance-priority recommendation
- Inspection-history visualization

Expected Prototype

A working platform demonstrating defect detection and maintenance prioritization using sample structure images or simulated sensor data.

KH26-CE-23 – Intelligent Construction Progress, Safety and Material-Waste Monitoring System

Background

Construction projects often face delays, unsafe work practices and excessive material wastage. Manual monitoring may not provide accurate real-time information to project managers.

Problem Statement

Develop an AI-enabled construction monitoring platform that analyses site images, work schedules and material records. The system should estimate project progress, identify safety violations and detect unusual material consumption. It should compare actual progress with planned milestones and generate alerts for delays or unsafe conditions. Project managers should be able to review site status, worker-safety observations and material-usage trends through a dashboard.

Mandatory Requirements

- Construction-progress tracking
- Planned-versus-actual comparison
- AI-based safety-violation detection
- Material-consumption monitoring
- Delay and wastage alerts
- Project-management dashboard

Expected Prototype

A functional application demonstrating progress, safety and material monitoring using sample construction-site data.

KH26-CE-24 – Smart Urban Water Distribution and Leakage Management System

Background

A significant amount of treated water is lost because of pipeline leakage, unauthorized usage and inefficient distribution. Water authorities often detect such problems only after visible damage or public complaints.

Problem Statement

Develop an intelligent water-distribution monitoring platform that analyses flow, pressure, storage and consumption data from different zones. The system should identify abnormal patterns, estimate possible leakage locations and recommend inspection priorities. It should monitor reservoir levels and suggest distribution plans during water shortages. Authorities must receive alerts for major leakage, excessive consumption and supply imbalance.

Mandatory Requirements

- Zone-wise water-flow monitoring
- AI-based leakage detection
- Possible leakage-location estimation
- Reservoir and demand tracking
- Supply-distribution recommendations
- Alert and analytics dashboard

Expected Prototype

A dashboard demonstrating leakage detection and distribution management using simulated pipeline-network data.

KH26-CE-25 – AI-Based Road Condition Assessment and Repair Prioritization Platform

Background

Road networks frequently develop potholes, cracks, surface deterioration and drainage-related damage. Maintenance decisions are often based on complaints rather than systematic condition assessment.

Problem Statement

Develop an AI-powered platform that analyses road images or video captured through mobile phones, vehicles or drones. The system should detect different types of road damage, estimate severity and associate each defect with its location. It should prioritize repairs based on damage level, traffic importance, accident risk and available maintenance budget. Authorities should be able to track reported, assigned and completed repair works.

Mandatory Requirements

- Road-image or video analysis
- Damage-type classification
- Severity and location identification
- Repair-priority calculation
- Budget-aware maintenance recommendation
- Repair-status tracking dashboard

Expected Prototype

A map-based application demonstrating road-damage detection, prioritization and maintenance tracking using sample images or video.